

1 DMC-PDV: Configuration in DMC

Purpose

As part of the basic PDV function, DMC sends inspection plans to the respective interface via a driver. DMC also processes incoming measured values, verifies limit values and sends the collected data and violated limit values to the HYDRA server.

Basic configuration of DMC components

The PDV connection includes two components:

- The component *HydraPdv* manages limit values and checks if limit values are violated.
- The component *HydraPdvWriter* writes limit values in a HYDRA-compatible format (HYDRA format) for processing in the server. Configure the *HydraPdvWriter* component as child of the *HydraPdv* component.

Use the *FileTransporter* to transfer data files generated by the *HydraPdvWriter* to a shared directory on the server (transaction safety).

The *HydraPdvWriter* includes the following configuration parameters:

- **FileTimespan:** specifies the frequency of generating a new data file for processing in the HYDRA server. Possible values:
 - o Month: a new file once a month.
 - o Day: a new file once a day.
 - o Hour: a new file once an hour.
- **OutputDirectory:** specifies a local directory that stores the data files. If you indicate a relative path, the directory is identified based on the current runtime directory (%ProgramData%\mpdv\mdc\[HUGO instance name]\...).

Standard *HydraPdvWriter* configuration:

```
<ComponentConfiguration Name="HydraPdvWriter"
  Class="mpdv.MachineDataCollector.Plugin.HydraPdv.HydraPdvFileWriter">
  <Parameter Name="FileTimespan" Value="Hour" />
  <Parameter Name="OutputDirectory" Value="PDV" />
</ComponentConfiguration>
```

The *HydraPdv* component includes the following configuration parameters:

- **ParameterMapping:** maps machine channels to characteristics. Enter data as follows: [Source entity]->[Parameter]=[Characteristic].

Standard *HydraPdvWriter* configuration:

```
<ComponentConfiguration Name="HydraPdv"
    Class="mpdv.MachineDataCollector.Plugin.HydraPdv.HydraPdvComponent"
    Children="HydraPdvWriter">
    <Parameter Name="ParameterMapping" Value="OPCDA->P:P002=EMG01" />
</ComponentConfiguration>
```

The *FileTransporter* component includes the following configuration parameters:

- **InputDirectory**: specifies the local directory that stores the files to be transferred (usually identical to the *HydraPdvWriter-OutputDirectory*). If you enter a relative path, the runtime directory is used as the basis.
- **OutputDirectory**: specifies a shared directory on the HYDRA server or of a network drive. This directory must be writable. You also have to change the HYDRA configuration in order to read PDV data files from this directory. If you enter a relative path, the runtime directory is used as the basis.
- **SearchPattern**: search dialog to restrict the files to be transferred. Use this dialog to define the file extensions that should be transferred by the *FileTransporter*. For *PDV* you can generally use the pattern "*.pdat|*.piat".
- **Interval**: interval in seconds at which DMC checks the input directory for new data files and, if necessary, transfers these files to the output directory.

Standard PDV *FileTransporter* configuration:

```
<ComponentConfiguration Name="HydraPdvTransporter"
    Class="mpdv.MachineDataCollector.Core.Components.FileTransporter">
    <Parameter Name="InputDirectory" Value="PDV" />
    <Parameter Name="OutputDirectory" Value="\\scc7\Hydra2\tmp\HugoPdvTransport" />
    <Parameter Name="Interval" Value="90" />
    <Parameter Name="SearchPattern" Value="*.pdat|*.piat" />
</ComponentConfiguration>
```

Basic configuration: limit values and monitoring

The *manufacturing instructions* include the defined target and limit values for every characteristic.

Example:

```
<ArrayOfProcessDataLimit xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <ProcessDataLimit>
        <Name>PointY</Name>
        <CheckLimits>true</CheckLimits>
        <UpperProcessLimit>575</UpperProcessLimit>
        <UpperToleranceLimit>565</UpperToleranceLimit>
        <TargetValue>550</TargetValue>
        <LowerToleranceLimit>535</LowerToleranceLimit>
        <LowerProcessLimit>525</LowerProcessLimit>
    </ProcessDataLimit>
</ArrayOfProcessDataLimit>
```

Parameters are defined as follows:

- **Name:** ID of the process parameter as specified in HYDRA.
- **UpperProcessLimit:** Upper process action limit; manual intervention is required if this value is exceeded.
- **UpperToleranceLimit:** Upper tolerance limit; manual intervention is optional if this value is exceeded.
- **TargetValue:** Target value of the process.
- **LowerToleranceLimit:** Lower tolerance limit; manual intervention is optional if this value is not reached.
- **LowerProcessLimit:** Lower process action limit; manual intervention is required if this value is not reached.
- **CheckLimits:** specifies if limit values should be checked or not. Possible values are *true* (check active) or *false* (check inactive).